BioOne RESEARCH EVOLVED

# Lower Triassic bivalve fossils from the Song Da and An Chau Basins, North Vietnam

Author(s): TOSHIFUMI KOMATSU and DANG TRAN HUYEN
Source: Paleontological Research, 11(2):135-144.
Published By: The Palaeontological Society of Japan
DOI: http://dx.doi.org/10.2517/1342-8144(2007)11[135:LTBFFT]2.0.CO;2
URL: http://www.bioone.org/doi/full/10.2517/1342-8144%282007%2911%5B135%3ALTBFFT%5D2.0.CO%3B2

BioOne (www.bioone.org) is a nonprofit, online aggregation of core research in the biological, ecological, and environmental sciences. BioOne provides a sustainable online platform for over 170 journals and books published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Web site, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/page/terms_of_use.

Usage of BioOne content is strictly limited to personal, educational, and non-commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Paleontological Research, vol. 11, no. 2, pp. 135–144, June 30, 2007
© by the Palaeontological Society of Japan

# Lower Triassic bivalve fossils from the Song Da and An Chau Basins, North Vietnam

TOSHIFUMI KOMATSU[1] AND DANG TRAN HUYEN[2]

[1]*Faculty of Science, Kumamoto University, Kumamoto 860-8555, Japan (e-mail: komatsu@sci.kumamoto-u.ac.jp)*

[2]*Department of Paleontology and Stratigraphy, Research Institute of Geology and Mineral Resources, Ministry of Industry, Vietnam (e-mail: rigmr@fpt.vn)*

Received February 28, 2006; Revised manuscript accepted December 7, 2006.

**Abstract. In North Vietnam, the Hua Tat, Pa Khom and Lang Son Formations yield abundant Lower Triassic bivalves. The occurrence of *Eumorphotis* sp. cf. *E. multiformis*, represents the first fossil reported from the Hua Tat Formation, while *Claraia aurita*, *C. concentrica*, and *C. intermedia multistriata* are found in the upper Lang Son Formation. The Pa Khom Formation contains *C. intermedia multistriata*, *Claraia* sp., and *Lopha* sp. The occurrence of *C. intermedia multistriata* indicates that the Pa Khom Formation and the upper Lang Son Formation are Olenekian in age. These *Claraia* spp. are found in wave-dominated nearshore deposits characterized by hummocky cross stratification and wave-ripple laminations, although *Claraia* in other areas is generally found in deep-sea mudstone.**

**Key words: *Claraia*, Lower Triassic bivalves, North Vietnam, wave-dominated environments**

## Introduction

The Paleozoic to Mesozoic systems are widely distributed in the Song Da and An Chau Basins, which extend further to North Vietnam and South China (Figure 1). These areas are important in terms of Triassic stratigraphy and Southeast Asian palaeobiogeography (Vu Khuc and Dang, 1998; Komatsu *et al*., 2006). The geology and stratigraphy of the An Chau Basin have been studied at many sites in the course of exploratory surveys for coal and bauxite deposits (Mansuy, 1908; Patte, 1926; Vu Khuc *et al*., 1991; Dang *et al*., 2004). The Lower Triassic Lang Son Formation occurs in eastern North Vietnam and contains many different Induan molluscan fossils (Vu Khuc *et al*., 1965; Vu Khuc, 1984).

In the Song Da Basin, Triassic deposits were first reported by Vu Khuc *et al*. (1965), and Dang *et al*. (2004) provided a detailed description of the Lower Triassic geology and stratigraphy (Figure 2). Molluscan fossils were also reported by Vu Khuc *et al*. (1991). According to Dang *et al*. (2004), the Lower Triassic strata in the Son Da Basin crop out in western North Vietnam, and are divided into two formations: the Hua Tat Formation, dominated by siliciclastic sedimentary rocks, and the Pa Khom Formation, composed of carbonates and siliciclastic rocks. The Pa Khom Formation yields Lower Triassic marine bivalves such as *Entolium discites microtis*, *Costatoria costata*, *Claraia* sp., and *Eumorphotis* sp. (Dang *et al*., 2004). Prior to our work there were no fossils recorded in the Hua Tat Formation, and, likewise, *Claraia* fossils were unknown in the lower Pa Khom Formation, and the upper Lang Son Formation at the Con Sang type section and Deo Lang section, Lang Son Province.

In this study, we describe some bivalve fossils from the Hua Tat, Pa Khom and Lang Son Formations, and discuss the age of these formations. We also reconstruct the depositional environments of fossil-bearing strata in the upper Hua Tat Formation, the lower Pa Khom Formation, and the upper Lang Son Formation, based on lithologies and sedimentary structures. Finally, we estimate the habitats of the characteristic Lower Triassic bivalves *Claraia*.

## Geologic setting and fossil localities

### Hua Tat and Pa Khom Formations in the Song Da Basin

The Lower Triassic Hua Tat and Pa Khom Formations crop out extensively in the Thanh Hoa, Hoa Binh, Ninh Binh, Son La and Lai Chau Provinces, western North Vietnam (Figure 1). The Triassic strata in this region strike NE-SE. The Hua Tat Formation

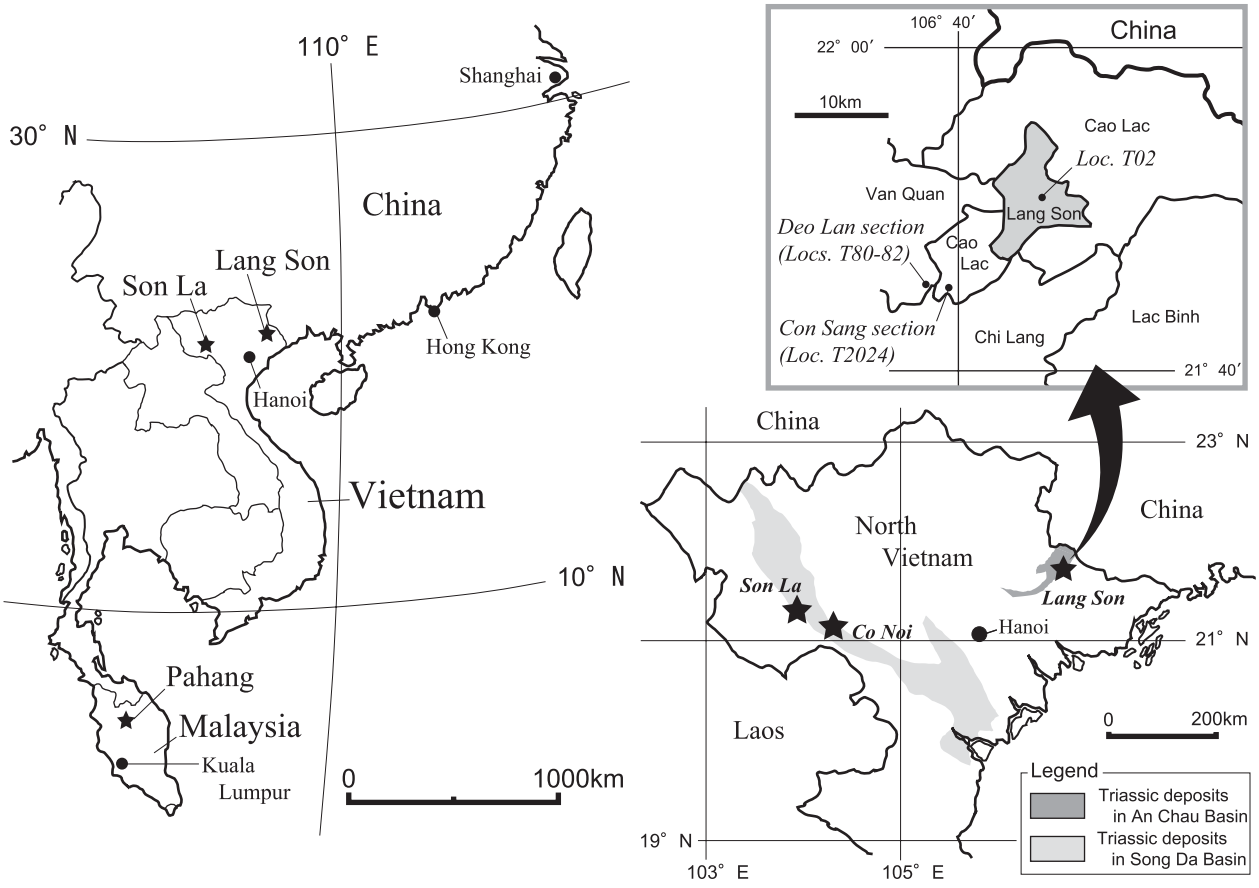



**Figure 1.** Index maps of the study area.

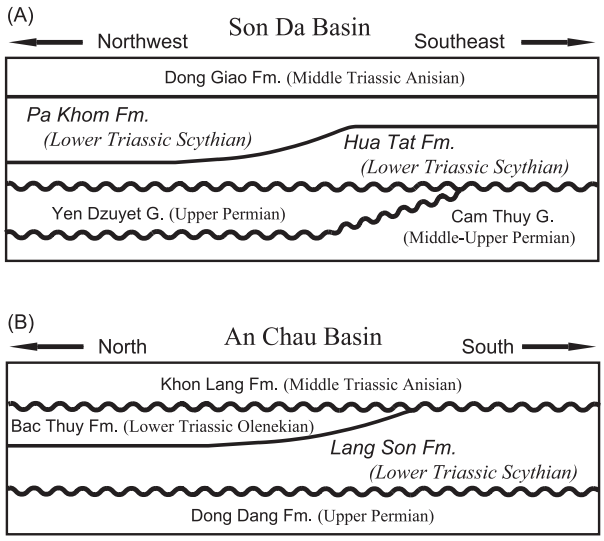



**Figure 2.** Stratigraphy of the Lower Triassic Hua Tat and Pa Khom Formations in Song Da Basin(A) and Lang Son Formation in An Chau Basin (B).

consists of pebble to cobble conglomerates, sandstone and mudstone, and unconformably overlies volcanic rocks of the Middle to Upper Permian Cam Thuy Group (Figure 2). In the lower part of the formation, conglomerate, gravelly sandstone and coarse- to medium-grained sandstone are dominant, while the upper part is mainly composed of fine- to very fine-grained sandstone and mudstone. The formation is 280 m thick at Quy Hau village in the southeast part of the basin, and thins rapidly to the northwest (Figure 2).

The Pa Khom Formation conformably overlies the Hua Tat Formation and comprises limestone, sandstone and mudstone that contain marine fossils such as bivalves, gastropods and ammonoids. The Lower Triassic (Olenekian) ammonoid *Meekoceras* sp. is found in the middle parts of this formation at the Co Noi type section (Dang *et al*., 2004). Although the Pa Khom Formation is 420 m thick at the type section, the thickness of the formation is variable in the north-

ern part of the basin, and it is 5 to 220 m thick in the Tua Chua Highland, Lai Chau Province.

In this study we collected bivalve fossils from various stratigraphic levels. In the upper Hua Tat Formation, *Eumorphotis* sp. cf. *E. multiformis*, is rarely found in mudstone at the Co Noi type section (Figures 3–5). In the lower Pa Khom Formation, Son La City, mudstone and very fine-grained sandstone yield gastropods, brachiopods and abundant epifaunal bivalves such as *Claraia intermedia multistriata*, *Claraia* sp., and *Lopha* sp. (Figures 3, 5). At the Co Noi type section, concentrations of thin shells (2–10 mm in bed thickness) are found in the lower and middle Pa Khom Formation. These shell concentrations are composed of poorly preserved small shells of *Claraia* sp. and *Eumorphotis* sp.

**Lang Son Formation in the An Chau Basin**

The Lower Triassic Lang Son Formation crops out in the Lang Son Province, eastern North Vietnam (Figure 1). The Lang Son Formation is about 200 m thick at the Con Sang type section and consists of sandstone, mudstone and calcareous mudstone. This formation unconformably overlies the Upper Permian Dong Dang Formation (Figure 2B) and is conformably overlain by limestone of the Lower Triassic (Olenekian) Bac Thuy Formation in Lang Son City (Thang, 1989). At the Con Sang type section, the Lang Son Formation is unconformably overlain by conglomerate and volcanic rocks of the Middle Triassic (Anisian) Khon Lang Formation (Dang and Nguyen, 2000).

Vu Khuc (1984) and Vu Khuc *et al*. (1965, 1991) described many Induan molluscan fossils in the Lang Son Formation, including *Claraia clarai*, *C. aurita*, *C. intermedia*, *C. stachei*, *C. wangi*, *C. vietnamica*, *C. kiparisovae*, *C. aurita* var. *gervilliformis*, *C. radialis*, and *C. kiparisovae* var. *brevis* (bivalves), *Glyptophiceras langsonense* and *Koninckites* sp. (ammonoids). The overlying Bac Thuy Formation yields abundant Olenekian ammonoids, including *Meekoceras* cf. *yukiangense*, *Meekoceras* sp., *Flemingites* sp. aff. *F. flemingianus*, *Owenites carinatus*, and *Ussuria lenticularis* (Vu Khuc, 1984), and contains Olenekian conodonts *Neospathodus dieneri*, *N. homeri*, *N. waageni*, and *N. triangularis* (Thang, 1989).

In Lang Son City, Lang Son Province, the upper Lang Song Formation is dominated by sandstone intercalated with mudstone beds, and yields abundant *Claraia concentrica* and *C. aurita* (Figure 3). In the Deo Lan area in central Lang Son Province, the upper Lang Song Formation is mainly composed of mudstone that contains *Claraia concentrica*, *C. aurita*, and *C. intermedia multistriata* (Figures 3, 4). Dang and Nguyen (2000) reported *Glyptophiceras* sp. (ammonoid), *Claraia stachei*, and *C. aurita* (bivalves) in the lower and middle Lang Son Formation from the Deo Lan area.

## Systematic paleontology

*Repository*.—Faculty of Science, Kumamoto University (KMSP). Faculty of Education, Kumamoto University (KE).

Order Pterioida Newell, 1965
Family Pseudomonotidae Newell, 1938
Genus ***Claraia*** Bittner, 1901

*Type species*.—*Posidonomya clarae* Emmrich, 1844

*Remarks*.—Ichikawa (1958) gave detailed revised diagnosis and descriptions of *Claraia*.

*Occurrence and age*.—The distribution of *Claraia* in Southeast Asia was compiled by Tamura (1968). Subsequent to this, abundant *Claraia* have been found at many other points in Asia (Nakazawa, 1977, 1981a; Chen *et al*., 1988; Chonglakmani and Grant-Mackie, 1993; Vu Khuc and Dang, 1998; Chen and Komatsu, 2001, 2002; Dang, 1998; Dang *et al*., 2004). In general, *Claraia* is regarded as a cosmopolitan species.

### ***Claraia concentrica*** (Yabe, 1928)

Figures 3.1–3.5

*Pseudomonotis* (*Aviculopecten*) *griesbachi* Bittner. Mansuy, 1908, p. 62, pl. 16, figs. 4–5.
*Pseudomonotis* (*Claraia*) *griesbachi* var. *concentrica* Yabe, 1928, p. 22–24, pl. 6, fig. 4; Hsu, 1937, p. 310, pl. 4, fig. 6; Chen, 1950, p. 89, pl. 1, fig. 9.
*Pseudomonotis* (*Claraia*) *wangi* Patte. Hsu, 1937, p. 312, pl. 1, fig. 6.
*Claraia aurita* var. *gervilliformis* Vu Khuc *et al*., 1965, p. 23, pl. 1, fig. 11.
*Claraia vietnamica* Vu Khuc *et al*., 1965, p. 24, 57, pl. 1, figs. 12–16.
*Claraia kiparisovae* Vu Khuc *et al*., 1965, p. 24, 57, pl. 1, figs. 17, 18, pl. 2, figs. 1, 2.
*Claraia kiparisovae* var. *brevis* Vu Khuc *et al*., 1965, p. 24, 57, pl. 2, figs. 3, 4.
*Claraia griesbachi concentrica* Yabe. Tamura, 1968, p. 80–81, pl. 11, figs. 1–14.
*Claraia concentrica* Yabe. Chen *et al*., 1974, p. 328, pl. 173, figs. 12, 13; Nakazawa, 1977, p. 201, pl. 3, figs. 9–11; Nakazawa, 1981a, p. 108–109, pl. 10, figs. 14–16.
*Claraia concentrica* (Yabe). Chen *et al*., 1988, pl. 1, figs. 2, 5.

*Materials*.—Holotype is a left valve collected at Tieh-shi-kou, Hsing-fen-hsien, Jiangxi Province, China (Yabe, 1928). In the Lang Son Formation, we collected blocks of shell concentrations dominated by *Claraia concentrica* (KMSP-2250; Figure 3.1). Left and right valves are shown (KMSP-2251; Figure 3.2).

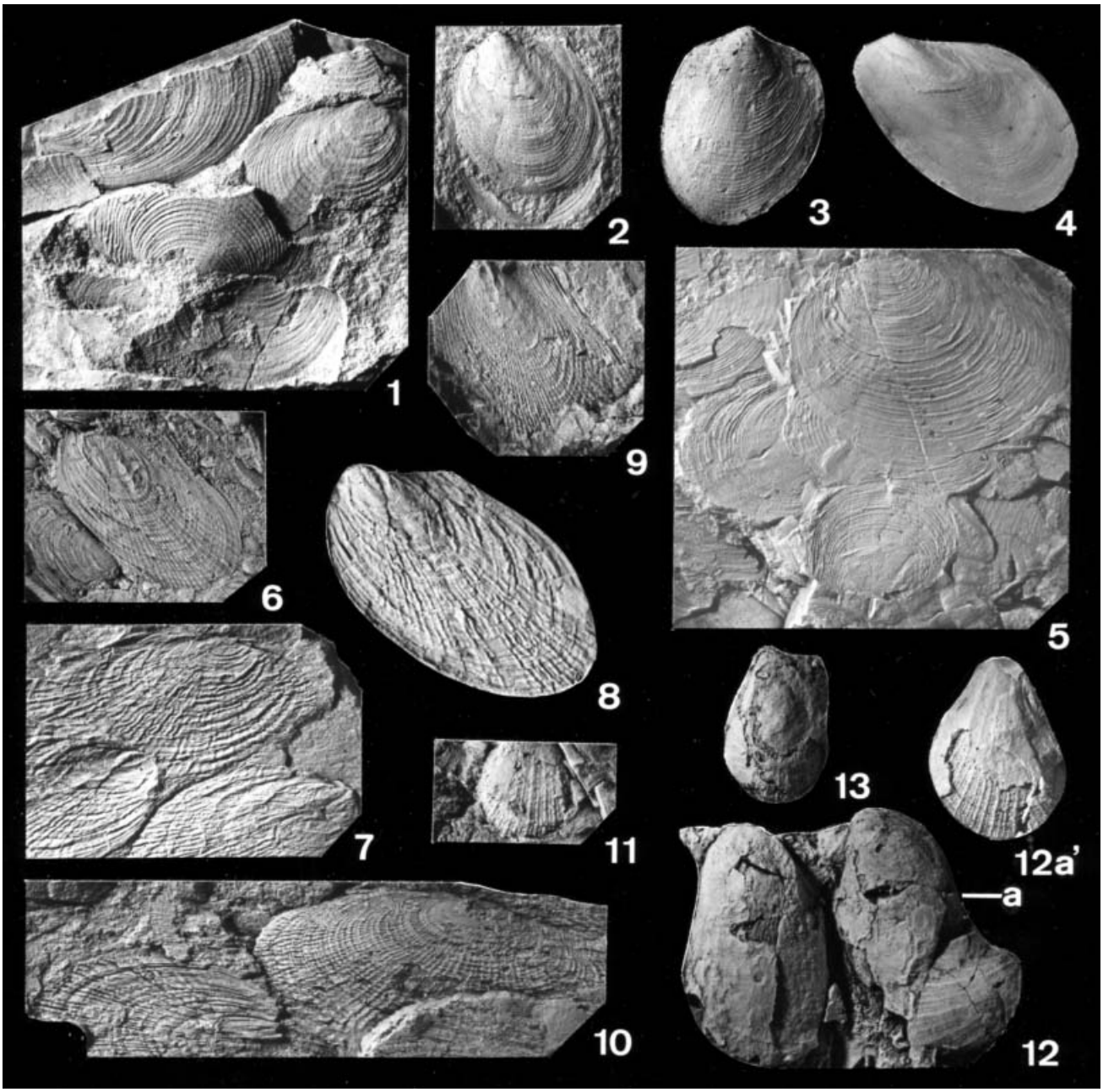



**Figure 3.** Lower Triassic bivalves. X1.0. KMSP = Registry number of Faculty of Science, Kumamoto University. KE = Registry number of Faculty of Education, Kumamoto University. **1–5**. *Claraia concentrica*. 1. Modes of occurrence of shell concentration, KMSP-2250, Lang Son Formation, Lang Son City, Lang Son Province (T02; 21°51′30″N, 106°44′28″E). 2. Left and right valves, KMSP-2251, Lang Son Formation, Con Sang, Lang Son province (T2024; 21°45′51″N, 106°38′37″E). 3. Holotype of *Claraia vietnamica*, No. 54g-1, Right valve, Lang Son Formation, Deo Sai Ho, Lang Son province. 4. Holotype of *Claraia aurita* var. *gervilliaeformis*, No. 54g-3, left valve, Deo Sai Ho, Lang Son province. 5. Mode of occurrence of shell concentration, rubber cast, KE1152 (Tamura, 1968, pl. 11, fig. 14), North Pahang, Malaya. **6–10**. *Claraia intermedia multistriata*. 6. Rubber cast of left valve, KE1157 (Tamura, 1968, pl. 11, fig. 17), North Pahang, Malaya. 7. Right valve, top parts of Lang Son Formation, Deo Lan, Lang Son province (T82; 21°45′32″N, 106°32′41″E). 8. Rubber cast of left valve, top parts of Lang Son Formation, Deo Lan, Lang Son province (T82; 21°45′32″N, 106°32′41″E). 9. Rubber cast of left valve, KMSP-2254, lower parts of Pa Khom Formation, Son La city, Son La province (Loc. 275; 21°19′54″N, 103°53′55″E). 10. Rubber cast of right valve, KMSP-2255, lower parts of Pa Khom Formation, Son La city, Song La province (Loc. 275; 21°19′54″N, 103°53′55″E). **11–13**. *Eumorphotis* sp. cf. *E. multiformis*, upper parts of Hua Tat Formation, Co Noi, Son La Province (T1743; 21°08′53″N, 104°11′51″E). 11. Rubber cast of left valve, KMSP-2256. 12. Modes of occurrence of shell concentration, KMSP-2257. 12a. Left internal mould. 12a′. Rubber cast of left valve. 13. Rubber cast of left valve, KMSP-2258.

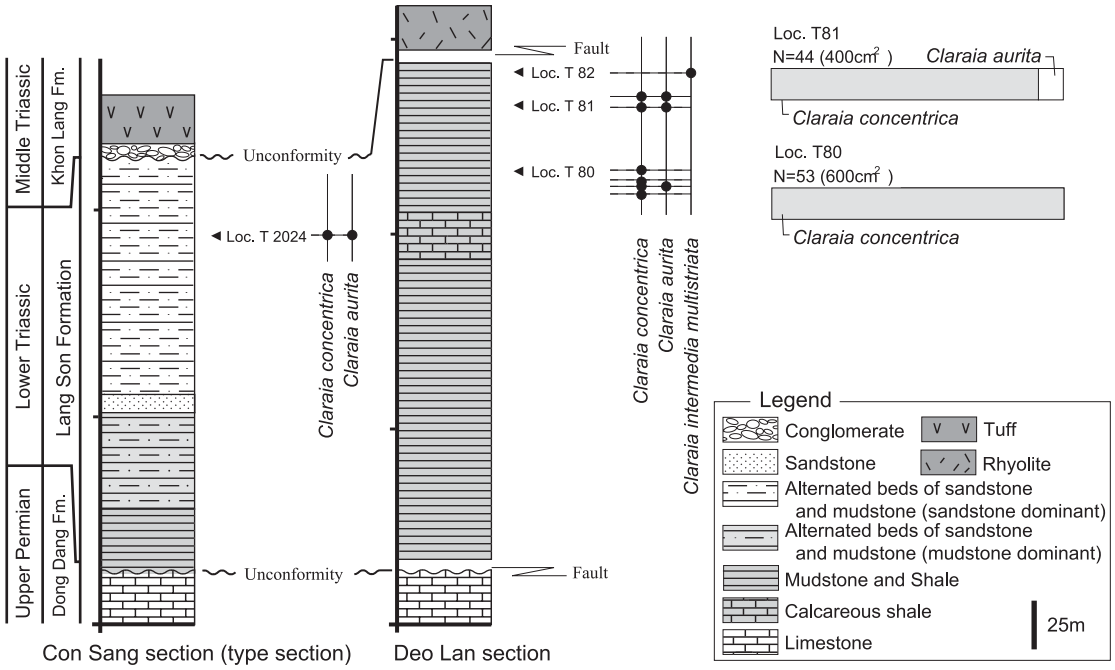



**Figure 4.** Columnar sections of the Lang Son Formation, showing the stratigraphic occurrences and compositions of bivalve fossils.

*Remarks*.—The holotype represents nearly a complete outline with clearly regular concentric ribs on the surface (Yabe, 1928). Tamura (1968) provided a detailed sketch of the species.

*Comparison*.—The specimens at hand are referable to *Claraia concentrica*, though some materials are characterized by much more narrowly interspaced fine regular concentric ribs (Figure 3.2). Similar morphotypes have been reported by Tamura (1968, pl. 11, figs. 8, 12) from North Pahang, Malaya. Some of the Malaya specimens have smooth and ridged fine concentric ribs, with interspaces much wider than the ribs (Figure 3.5).

Mansuy (1908) reported *Pseudomonotis griesbachi* Bittner (= *Claraia concentrica*) from the Lower Triassic deposits in Dang-quan, Tonkin; these deposits are probably correlatable with the Lang Son Formation in Lang Son Province, North Vietnam. Vu Khuc *et al*. (1965) described *C. vietnamica*, *C. kiparisovae*, *C. aurita* var. *gervilliformis*, and *C. kiparisovae* var. *brevis* from the Lang Son Formation, Deo Sai Ho area, Lang Son Province. However, *C. vietnamica*, *C. kiparisovae*, *C. aurita* var. *gervilliformis*, and *C. kiparisovae* var. *brevis* are sometimes strongly deformed, because the fossil-bearing strata are complexly faulted and folded (Figures 3.3–3.4). These materials are characterized by wide or narrow interspaced fine regular concentric ribs, although the shell outlines are variable in shape. For example, the shape of *C. vietnamica* is a vertically elongate oval (Figure 3.3), and *C. kiparisovae* and *C. kiparisovae* var. *brevis* are laterally elongate oval in shape. These deformed materials are probably related to *C. concentrica*.

*Claraia concentrica* is abundant in Lower Triassic deposits of South and Southeast Asia. The types were obtained from Tieh-shi-kou and Hsing-fen-hsien, Jiangxi Province, South China (Yabe, 1928). Tamura (1968) collected abundant *C. griesbachi concentrica* (= *C. concentrica*) from northern Pahang, North Malaya. Furthermore, Nakazawa (1981b) proposed the *C. concentrica* Biozone ranging from Late Griesbachian to Early Dienerian in the lower Khunamuh Formation, Kashmir, India. *C. concentrica* has not been found in Japan and eastern Russia, although some other *Claraia* (ex *C. aurita* and *C. pulchella*) have been reported from these areas (Nakazawa, 1953; Tamura, 1968). *Claraia concentrica* is probably a characteristic Triassic Tethyan species in South and Southeast Asia.

*Occurrence*.—*Claraia concentrica* is abundant in the Lang Son Formation, Lang Son City, Lang Son Province (Loc. T02; 21°51′30″N, 106°44′28″E) and the Con Sang area, Lang Son Province (Loc. T2024; 21°45′51″N, 106°38′37″E).

***Claraia intermedia multistriata*** Ichikawa in Ichikawa and Yin, 1966

Figures 3.6–3.10

*Pseudomonotis intermedia* Bittner. Wittenburg, 1908, p. 26, pl. 3, figs. 3–5.
*Claraia intermedia* (Bittner). Leonardi, 1935, p. 60, pl. 3, figs. 5, 8.
*Claraia intermedia multistriata* Ichikawa in Ichikawa and Yin, 1966, p. 102–104, pl. 2, figs. 1–6; Tamura, 1968, p. 81, pl. 11, figs. 15–17.

*Remarks on the holotype*.—Ichikawa in Ichikawa and Yin (1966) proposed *Claraia intermedia multistriata* on the basis of *Pseudomonotis intermedia* Bittner of Wittenburg (1908). The holotype was collected from the Lower Triassic Seis Formation of the Werfen Group distributed in Campitello (Durontal) of the Dolomite Alps (Wittenburg, 1908). The present subspecies is characterized by densely set uniform radial striae, a well developed posterior wing, and numerous regular concentric ribs. Although *C. intermedia multistriata* is similar to *Claraia intermedia intermedia*, the radial striae of the former are clearly coarser and stronger than those of the latter.

*Material*.—One left valve and two right valves were collected from the upper Lang Son Formation (KMSP-2254, 2255; Figures 3.9–3.10). Abundant specimens occur in the Pa Khom Formation, and one left valve and one right valve are shown (KMSP-2252, 2253; Figures 3.7–3.8).

*Descriptive remarks*.—Shell large for genus, inflated, inequivalve and inequilateral, prosocline, suborbicular in outline; test thin; left valve slightly convex; posterior wing well developed for genus, its posterior margin slightly compressed; umbo slightly protruded above hinge margin, situated about 1/3 from anterior margin; hinge line long and straight; anterior wing small without sinus, anterior, posterior and ventral margins rounded; surface ornamented with densely arranged straight radial striae and numerous regular fine concentric ribs; both ribs and ridges smooth; both interspaces of ribs much wider than ribs. Right valve slightly convex; anterior wing short but distinct; byssal sinus deep and wide; other characteristics similar to those of left valve.

*Comparisons*.—In spite of being deformed and fragmented, diagnostic features such as well developed posterior wing, densely set uniform coarse radial striae, and numerous regular concentric ribs are well preserved. The coarse concentric ribs clearly cover umbonal to ventral areas. In right valves, anterior wings are characterized by a distinct and deep byssal sinus (Figure 3.7) and are morphologically similar to the anterior wings of Malayan species from northern Pahang and southern Kelantan, North Malaya, as described by Ichikawa and Yin (1966, pl. 2, fig. 5).

*Occurrence*.—The present species is abundant in very fine-grained sandstone and mudstone of the lower Pa Khom Formation, Son La City, Son La Province (Loc. 275; 21°19′54″N, 103°53′55″E) and shale of the upper Lang Son Formation, Deo Lan area, Lang Son Province (T82; 21°45′32″N, 106°32′41″E).

Family Aviculopectinidae Meek and Hayden, 1964
Genus ***Eumorphotis*** Bittner, 1901

Type species.—*Pseudomonotis telleri* Bittner, 1899

***Eumorphotis* sp.** cf. ***E. multiformis*** Bittner

Figures 3.11–3.13

*Material*.—Six specimens, three rubber casts from left external moulds (KMSP-2256, 2257, 2258; Figures 3.11, 3.12a′, 3.13), and three internal moulds (KMSP-2257; Figure 3.12).

*Comparisons*.—These materials are strongly deformed, and shell ornamentations and the umbonal area of some materials are only partly preserved. However, characteristic first-, second- and third-order ribs with finer radial striae (= fourth rib in Bittner, 1899) are observable. Occasionally, the second-order ribs are of similar strength to the third-order ones. Generally, about four ribs of the second and third orders are arranged in the interspaces of the first-order ribs, which represent the characteristic ornament of *E. multiformis*. The anterior wing is trigonal, sharply separated from the disc by the sinus. The posterior wing is laterally elongate trigonal, and is a little larger than the anterior wing. The posterior wing is not clearly separated from the disc, and shows a rounded posterior end. These dorsal characters are also similar to *E. multiformis*.

*Occurrence*.—*Eumorphotis multiformis* has been reported from Europe, North America, Russia, South China, India, and Japan (Bittner, 1899; Newell and Kummel, 1942; Yin, 1981; Nakazawa, 1981a; Dang, 1998), and is a cosmopolitan species (Hallam and Wignall, 1997). In North Vietnam, this species is found in the Lower Conoi Formation (= Pa Khom Formation) and the Hong Ngai Formation, Hong Ngai area, Dong Van, Ha Giang Province (Vu Khuc *et al*., 1991; Dang, 1998).

Rare examples of the present species are found in mudstone of the upper Hua Tat Formation, Co Noi, Son La Province (T1743; 21°08′53″N, 104°11′51″E).

## Depositional environments of fossil-bearing strata and bivalve habitat

### Hua Tat and Pa Khom Formations of the Song Da Basin

Lithology and sedimentary features of the upper Hua Tat Formation are clearly different from those of the lower Pa Khom Formation. The upper Hua Tat Formation consists of sandstone and mudstone (Figure 5). These siliciclastic deposits are characterized by lenticular, wavy and flaser bedding 0.1–10 m thick intercalated with thin massive mudstone and thick bioturbated mudstone that yields *Eumorphotis* sp., cf. *E. multiformis*. The lenticular, wavy and flaser beds (0.1–10 m in thickness) contain current and climbing ripple-laminations. In some beds, the forests of current ripples occasionally represent bidirectional dipping; ripple-laminations indicative of bimodal currents probably developed in a tide-dominated environment. The lenticular, wavy and flaser bedded sandstones and mudstones are interpreted to represent deposition in a tidal flat environment. Bioturbated mudstone was probably deposited in an enclosed bay or lagoon that faced tidal flats; the bioturbation represents the activity of benthic animals within the sediment or on the sediment surface.

The lower Pa Khom Formation is composed of limestone, sandstone and mudstone (Figure 5): cross-stratified sandstone and gravelly sandstone is commonly found. In particular, very fine-grained sandstone at Quy Hau village, Hoa Binh Province, occasionally exhibits hummocky cross-stratification (HCS) (Figures 5, 6). At the Co Noi type section and Son La City (Locs. T275, T276), lenticular and wavy beds 1–7 m thick consist of wave- and current-ripple laminated sandstone and massive mudstone. Generally, wave-ripple laminations (10–20 cm in wave length) are characterized by chevron and off-shoot structures, and are dominant in lenticular and wavy beds. HCS is a sedimentary structure diagnostic of waning storm and wave conditions or combined flow (e.g., Walker and Plint, 1992; Cheel and Leckie, 1993). Therefore, these deposits are strongly influenced by wave and storm currents; this likely indicates that sandstone in the lower Pa Khom Formation accumulated in a shoreface environment.

In the lower Pa Khom Formation, wavy- and flaser-bedded sandstone and mudstone yield bivalves such as *Claraia intermedia multistriata*, *Claraia* sp. and *Lopha* sp. (Locs. T275, T276; Figure 5). These fossils form thin shell concentrations composed mainly of disarticulated *Claraia*. Occasionally, small *Claraia* sp. (2–6 mm in shell length) form monospecific shell concentrations. Besides, articulated *Claraia* is commonly found at loc. T276. At this locality, extremely thin and frangible shell remains of *Claraia* are nonabraded and still keep detailed growth lines and fine radial ribs. The nature of shell preservation and mode of occurrence indicates a parautochthonous origin, suggesting

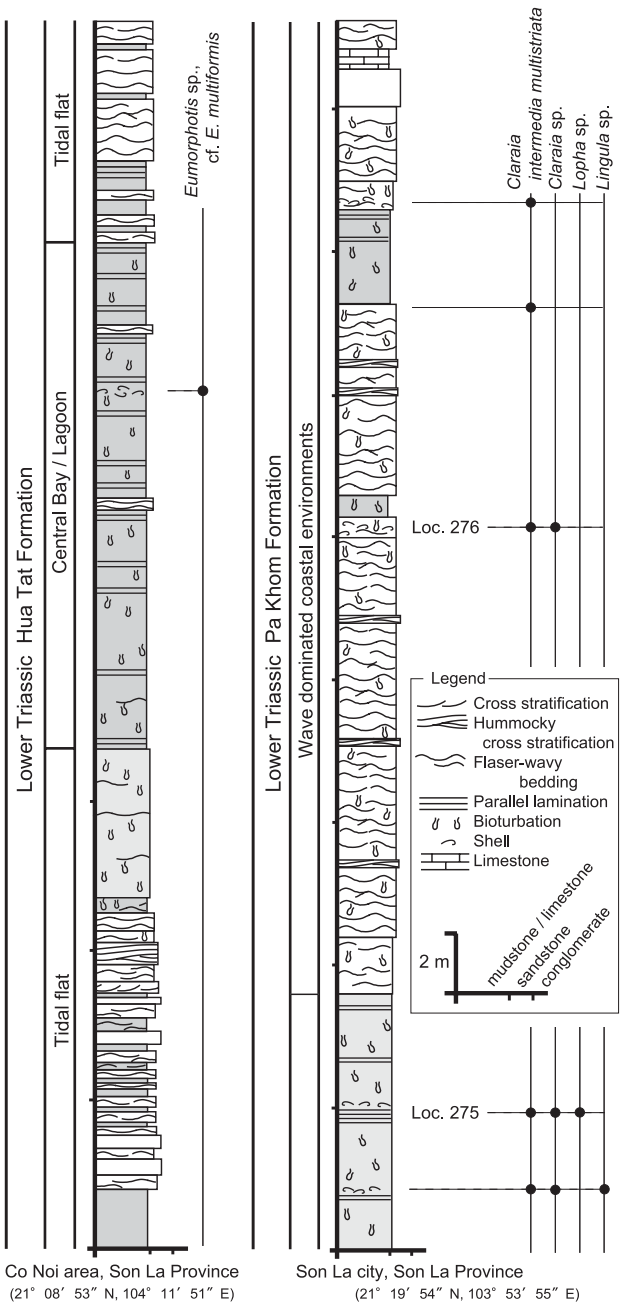



**Figure 5.** Columnar sections of the upper part of the Hua Tat and Pa Khom Formations, showing the stratigraphic occurrences of bivalve fossils.

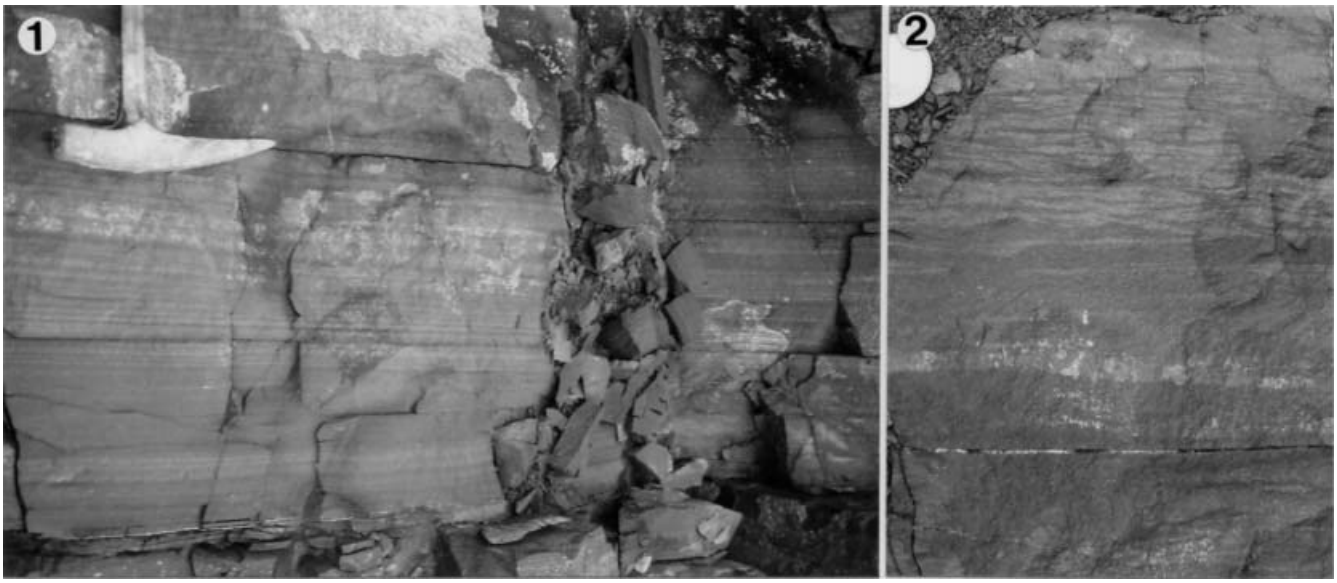



**Figure 6.** Photos showing sedimentary structures. **1**. Hummocky cross-stratified sandstone in the Pa Khom Formation (Quy Hau, Hoa Binh Province; 20°38′11″N, 105°16′45″E). **2**. Flaser and wavy bedding containing wave and current ripple laminations in the Pa Khom Formation (Co Noi, Son La Province; 21°08′53″N, 104°11′51″E).

that *C. intermedia multistriata* inhabits wave- and storm-dominated shoreface environments. *Eumorphotis* sp. cf. *E. multiformis* in the Hua Tat Formation is strongly abraded and fragmented; its occurrence is typical of an allochthonous origin.

**Lang Son Formation in the An Chau Basin**

At Lang Son City (Loc. T02), Lang Son Province, the upper Lang Son Formation is composed of alternating beds of very fine-grained sandstone and mudstone. Sandstone beds are 3–60 cm in thickness and characterized by HCS and wave-ripples. Climbing ripple laminations are rarely found in thinly flaser-bedded sandstone. Mudstone beds are 2–20 cm in thickness and contain parallel-laminations and trace fossils.

Thin shell layers, 5 mm in thickness, occasionally occur on the upper surfaces of HCS sandstone beds. The concentrations are mainly composed of fragments and disarticulated valves of *Claraia concentrica* and *C. aurita*, indicating typical allochthonous occurrences in a wave- and storm-dominated shoreface and shelf environments.

## Geological ages

*Claraia* is an important Lower Triassic age indicator (Nakazawa, 1981b; Yin, 1981; Ogg, 2004) and is abundant in the lower Pa Khom Formation and the upper Lang Son Formation. *Claraia concentrica* is reported from South China, North Malaya and Kashmir, India (Yabe, 1928; Tamura, 1968; Nakazawa, 1981a), and indicates an Induan to Early Olenekian age (Tamura, 1968).

Nakazawa (1981a, b) reported a *Claraia concentrica* Biozone in the Kashmir area with an Induan (Late Griesbachian to Early Dienerian) age. This bivalve zone is equivalent to the upper half of the *Neogondolella carinata carinata* conodont zone, the lower half of the *Neospathodus cristgalli* zone, and the upper half of the *Ophiceras* ammonoid zone (Nakazawa, 1981b). These conodont and ammonoid zones are generally Induan in age (Ogg, 2004).

The European *Claraia intermedia multistriata* is rarely found in the Induan Seis Formation of the Werfen Group in the Dolomite Alps (Wittenburg, 1908; Leonardi, 1935; Ichikawa and Yin, 1966); however, this species is abundant in the Lower Triassic strata of North Malaya (Ichikawa and Yin, 1966; Tamura, 1968). The range of *C. intermedia multistriata* is reckoned as Early Olenekian (= Smithian) in a figure showing the range chart of *Claraia* in Southeast Asia (Tamura, 1968). In Vietnam, *C. intermedia multistriata* is found in the uppermost Lang Son Formation (Loc. T82) and the lower Pa Khom Formation (Loc. 275). In the Pa Khom Formation, *Entolium discites microtis*, *Costatoria costata* (bivalves) and *Meekoceras* sp. (ammonoid) indicate an Olenekian age (Dang *et al*., 2004).

The Pa Khom Formation and the upper Lang Son Formation are therefore probably Olenekian in age.

The upper Hua Tat Formation yields *Eumorphotis* sp. cf. *E. multiformis*. Generally, *E. multiformis* is abundant in the Lower Triassic, mainly Induan, shallow marine deposits (Yin, 1981; Dang, 1998). In Japan, the lower Kamura Formation, which is subdivided into several conodont zones (Koike, 1996), intercalates shell beds that are mainly composed of *E. multiformis*. In particular, Dienerian (= Upper Induan) and Smithian (= Lower Olenekian) limestone that contains *Neogondolella dieneri* and *N. conservatives* (conodonts) yields abundant *E. multiformis*. On the other hand, the present species are not found in the Lower Griesbachian limestone, which is characterized by conodonts such as *Hindeodus parvus* and *Neogondolella carinata*. *E. multiformis* is reported from Upper Griesbachian strata in Kashmir, but not from Lower Griesbachian strata characterized by occurrence of *Otoceras* (ammonoid) (Nakazawa, 1981a, b). Shallow marine deposits in the upper Hua Tat Formation probably accumulated during the Late Induan.

## Conclusions

The Hua Tat Formation is characterized by tidal flat and lagoon/bay deposits, and yields *Eumorphotis* sp. cf. *E. multiformis. Claraia intermedia multistriata* is commonly found in the lower Pa Khom Formation and the uppermost Lang Son Formation, and indicates an Olenekian age. Besides, articulated and well preserved *Claraia aurita*, *Claraia concentrica* and *C. intermedia multistriata* are abundantly found in wave- and storm-dominated shoreface sandstone and muddy shelf deposits, which probably suggest that their habitats represent shoreface to shelf environments.

The study of Lower Triassic bivalve assemblage remains largely to be explored (Aberhan, 1994; Komatsu *et al.*, 2006). However, in North Vietnam, widely distributed Lower Triassic deposits abundantly yield well preserved fossils. These fossils and successive stratigraphic data are important for paleoecology, biostratigraphy, and paleobiogeography in South East Asia, and possibly provide keys and answers to difficult questions about correlations.

## Acknowledgements

We are grateful to Haruyoshi Maeda (Kyoto University) for his critical reading of the manuscript. Komatsu wishes to thank the staff of the Research Institute of Geology and Mineral Resources (RIGMR), Vietnam for their valuable advice and kind help. We also thank Vo Xuan Dinh (RIGMR) for his help in our field survey. This study was financially supported by a Grant-in-Aid for Encouragement of Young Scientists from the Ministry of Education, Science and Culture of the Government of Japan (no. 40336201).

## References

Bittner, A., 1899: Beiträge zur Paläontologie, insbesondere der triadischen Ablagerungen centralasiatischer Hochgebirge. *Jahrbuch der kaiserlich-königlichen Geologischen Reichsanstalt*, vol. 48, p. 689–718.

Bittner, A., 1901: Über *Pseudomonotis telleri* und verwandte Arten der unteren Trias. *Jahrbuch der kaiserlich-königlichen Geologischen Reichsanstalt*, vol. 50, p. 559–592.

Cheel, R. J. and Leckie, D. A., 1993: Hummocky cross-stratification. *Sedimentology Review*, vol. 1, p. 103–122.

Chen, C. C., Ma, G. H. and Chang, Z. M., 1974: Triassic Bivalvia. *In*, Nanjing Institute of Geology and Palaeontology, Academia Sinica *eds*., *A handbook of the Stratigraphy and Palaeontology in South-west China*. p. 326–343. (*in Chinese*)

Chen, C. Z., Wang, Y. G., Wang, Z. H. and Huang, P., 1988: Triassic biostratigraphy in South Jiangsu. *In*, Academy of Geological Sciences, Jiangsu Bureau of Petroleum Prospecting and Nanjing Institute of Geology and Palaeontology, Academia Sinica *eds*., *Sinian-Triassic Biostratigraphy of the Lower Yangtze Peneplatform in Jiangsu Region*, p. 315–368. Nanjing University Press, Nanjing. (*in Chinese*)

Chen, J. H. and Komatsu, T., 2001: Bivalves from the beds near the Permian-Triassic boundary at Meishan section of Changxing, south China and the correlation with shallow marine bivalve fauna. *Proceeding of the International Symposium on the Global Stratotype of the Permian-Triassic Boundary and the Paleozoic-Mesozoic Events*, p. 37–39.

Chen, J. H. and Komatsu, T., 2002: So-called Middle Triassic "*Claraia*" (Bivalvia) from Guangxi, South China. *Acta Palaeontologica Sinica*, vol. 41, p. 434–447.

Chen, P. Y., 1950: On the marine Triassic fauna of Lungyen and Ning Yang, Fukien. *Quarterly Journal of the Taiwan Museum*, vol. 3, p. 82–99.

Chonglakmani, C. and Grant-Mackie, J. A., 1993: Biostratigraphy and facies variation of the marine Triassic sequences in Thailand. *International Symposium on Biostratigraphy of Mainland Southeast Asia: Facies & Paleontology*, vol. 1, p. 97–123.

Dang, T. H., 1998: Lower Triassic bivalves from the Hong Ngai Formation (Song Hien Structural Zone). *Journal of Geology, Series B*, no. 11–12, p. 95–106.

Dang, T. H., Nguyen, D. H., Nguyen, H. H., Doan N. T. and Nguyen, L. N., 2004: New data on stratigraphy and paleontology of Lower Triassic sediments in West Bac Bo. *Journal of Geology, Series A*, no. 283, p. 1–9. (*in Vietnamese with English abstract*)

Dang, T. H. and Nguyen, K. Q., 2000: Stratigraphical and paleontological data on Lower Triassic sediments in the An Chau structure facies zone. *Geology and Mineral Resources*, vol. 7, p. 9–24. (*in Vietnamese with English abstract*)

Emmrich, H., 1844: Über die Schichten-Folge der Flötz-

Gebirge des Gader-Thales, der Seisser-Alpe und insbesondere bei St. Cassian. *Neues Jahrbuch für Mineralogie Geognosie Geologie und Petrefaktenkunde*, p. 790–803.

Hallam, A. and Wignall, P. B., 1997: *Mass extinctions and their aftermath*, 320 p. Oxford University Press, Oxford, New York.

Hsu, T. Y., 1937: Contribution to the marine Lower Triassic fauna of southern China. *Bulletin of the Geological Society of China*, vol. 16, p. 303–346.

Ichikawa, K., 1958: Zur Taxionomie und Phylogenie der triadischen "Pteriidae" (Lamellibranch.), mit besonderer Berücksichtigung der Gattungen *Claraia*, *Eumorphotis*, *Oxytoma* und *Monotis. Palaeontographica, Abteilung A*, Bd. 111, p. 131–212.

Ichikawa, K. and Yin, E. H., 1966: Discovery of Early Triassic bivalves from Kelantan, Malaya. *Journal of Geosciences, Osaka City University*, vol. 9, p. 101–106.

Koike, T., 1996: The first occurrence of Griesbachian conodonts in Japan. *Transactions and Proceedings of the Palaeontological Society of Japan, New Series*, no. 181, p. 337–346.

Komatsu, T., Dang, T. H. and Chen, J. H., 2006: Stratigraphy, depositional environments and bivalve assemblages in the Lower Triassic System in North Vietnam and South China. *Journal of Geography*, vol. 115, p. 470–483. (*in Japanese with English abstract*)

Leonardi, P., 1935: Il Trias inferiore delle Venezie. *Memorie dell'Istituto Geologico della R. Università di Padova*, vol. 11, p. 1–92. (*in Italian*)

Mansuy, H., 1908: *Contribution à la carte géologique de l'Indochine*, 73p. Imprimerie d'Extréme-Orient, Ha Noi.

Nakazawa, K., 1953: Discovery of *Claraia* and *Eumorphotis* from Triassic Yakuno Group, Kyoto Pref. Japan. *Memoirs of the College of Science University of Kyoto, Series B*, vol. 20, p. 261–269.

Nakazawa, K., 1977: On *Claraia* of Kashmir and Iran. *Journal of the Palaeontological Society of India*, vol. 20, p. 191–204.

Nakazawa, K., 1981a: Permian and Triassic bivalves from Kashmir. *In*, Nakazawa, K. and Kapoor, H. M. *eds*., The Upper Permian and Lower Triassic faunas of Kashmir. *Memoirs of the Geological Survey of India, Palaeontologia Indica*, new series, vol. 46, p. 87–122.

Nakazawa, K., 1981b: Analysis of Late Permian and Early Triassic faunas of Kashmir. *In*, Nakazawa, K. and Kapoor, H. M. *eds*., *Memoirs of the Geological Survey of India, Palaeontologia Indica*, vol. 46, p. 191–204.

Newell, N. D. and Kummel, B., 1942: Lower Eo-Triassic stratigraphy, Western Wyoming and Southeast Idaho. *Bulletin of the Geological Society of America*, vol. 53, p. 937–996.

Ogg, J. G., 2004: The Triassic Period. *In*, Gradstein, F. M., Ogg, J. G. and Smith, A. G. *eds*., *A Geologic Time Scale 2004*, p. 271–306. Cambridge University Press.

Patte, E., 1926: Fossiles paléozoiques et mésozoiques du Sud-Ouest de la Chine. *Palaeontologia Sinica, Series B*, vol. 15, p. 1–42.

Thang, B. D., 1989: Lower Triassic conodonts from North Vietnam. *Acta Palaeontologica Polonica*, vol. 34, p. 391–416.

Tamura, M., 1968: *Claraia* from North Malaya, with a note on the distribution of *Claraia* in Southeast Asia. *Geology and Paleontology of Southeast Asia*, vol. 5, p. 78–87.

Vu Khuc, 1984: *Triassic ammonoids in Vietnam*. 136 p. General Department of Geology of SR Vietnam, Hanoi. (*in Vietnamese*)

Vu Khuc, Dagyx, A. X., Kiparixova, L. D., Nguyen, B. N., Truong, C. A. and Srebrodolskaia, I. N., 1965: *Characteristic fossils of Triassic of North Vietnam*, 118 p. General Department of Geology of DR Vietnam, Hanoi. (*in Vietnamese*)

Vu Khuc and Dang T. H., 1998: Triassic correlation of the Southeast Asian mainland. *Palaeogeography, Palaeoclimatology, Palaeoecology*, vol. 143, p. 285–291.

Vu Khuc, Vu Chau, Trinh D., Dang, T. H., Nguyen, D. H. and Trinh, T., 1991: *Paleontological Atlas of Vietnam*, *vol. 3, Mollusca*, 207 p. Science and Technics Publishing House, Hanoi.

Walker, R. G. and Plint, A. G., 1992: Wave and storm-dominated shallow marine systems. *In*, Walker, R. G. and James, N. P. *eds*., *Facies Models: Response to Sea Level Change*, p. 219–238. Geological Association of Canada, Stittsville, Ontario.

Wittenburg, P. V., 1908: Beiträge zur Kenntnis der Werfener Schichten Süditirols. *Geologische und Paläontologische Abhandlungen*, vol. 12, p. 1–44.

Yabe, H., 1928: Notes on the interesting fossils from South China. *Japanese Journal of Geology and Geography*, vol. 6, p. 19–25.

Yin, H., 1981: Palaeogeographical and stratigraphical distribution of the Lower Triassic *Claraia* and *Eumorphotis* (Bivalvia). *Acta Geologica Sinica*, vol. 3, p. 161–169.